

In [4]:

```
# 1. sum() and multiply() func
def sum(l):
    s=0
    for i in l:
        s+=i
    return s

def multiply(l):
    m=1
    for i in l:
        m*=i
    return m

print(f'The sum of the elements: {sum([1,2,3,4])}')
print(f'The product of the elements: {multiply([1,2,3,4])}')
```

The sum of the elements: 10
The product of the elements: 24

In [5]:

```
# 2. display max, min, 2nd max
def displayMinMax(l):
    print(f'max: {max(l)}')
    print(f'min: {min(l)}')

def displaySecondMax(l):
    for i in l:
        if i==max(l):
            l.remove(i)
    print(f'The second max: {max(l)}')
```

In [6]:

```
# 3. linear search
def linearSearch(l, key, start, end):
    if start > end:
        return -1

    if l[start] == key:
        return start

    return linearSearch(l, key, start + 1, end)

l = [1, 2, 3, 4, 5]
key = 3
pos = linearSearch(l, key, 0, len(l) - 1)

if pos == -1:
    print("The key element was not found.")
else:
    print("The key element was found at index:", pos)
```

The key element was found at index: 2

In [14]:

```
# 4. Binary Search
def binarySearch(l,k):
    beg=0
    end=len(l)-1
    while(beg<=end):
        mid=(beg+end)//2
        if l[mid]==k:
            return mid
        elif l[mid]>k:
            end=mid-1
        elif l[mid]<k:
            beg=mid+1
    else:
        print('element not found')

l=[10,20,30,40,50]
print(f'the elemnet is found at pos: {binarySearch(l,20)+1}')
```

the elemnet is found at pos: 2

In [19]:

```
# 5. Fibonacci series
def fibonaci(n):
    f=[0,1]
    for i in range(2,n):
        f.append(f[i-1]+f[i-2])
    return f

n=int(input('Enter a no. of terms in fibonacci'))
print(f'The first {n} fibonacci series are \n{fibonaci(n)} ')
```

The first 10 fibonacci series are
[0, 1, 1, 2, 3, 5, 8, 13, 21, 34]

In [2]:

```
# 6. Soln of quadratic eqn
import math
def findQuadraticRoots(a,b,c):
    d=b**2-4*a*c
    if d>0:
        r1=(-b+d**0.5)/(2*a)
        r2=(-b-d**0.5)/(2*a)
        print(f'the roots are {r1} and {r2}')
    elif d==0:
        r=-b/(2*a)
        print(f'both the roots are {r}')
    else:
        x = -b / (2*a)
        y = math.sqrt(-d) / (2*a)
        print("The quadratic equation has two complex roots:")
        print("Root 1 = ", x, "+", y, "i")
        print("Root 2 = ", x, "-", y, "i")

findQuadraticRoots(1,-4,4)
```

both the roots are 2.0

In [11]:

```
sub=['numerical techniques','DCN','DAA','TOC','data science','DAA-Lab','web technolo
print('To display all the elements of the list using loop:')
for i in sub:
    print(i)
print(f'\nThe 2nd and 5th subjects are "{sub[1]}" and "{sub[4]}" respectively\n')
print(f'The first 4 subjects are\n{sub[:4]}\n')
print(f'The last 4 subjects are\n{sub[-4:]}\n')
print(f"'Python-Lab' in sub\n")
#append and insert
sub.append('COI')
print(f'the list after appendng COI is: \n{sub}\n')
sub.insert(0,'unix')
print(f'the list after inserting unix at pos(0) is: \n{sub}\n')
#pop and remove
sub.pop()
print(f'the list after pop() is: \n{sub}')
sub.remove('unix')
print(f'the list after removing "unix" is: \n{sub}')
#extend
sub.extend(['AD41','AD42','AD43','AD44','AD45','ADL46','ADL47','ADL48'])
print(sub)
#ascending and decending order
sub.sort()
print(sub)
sub.sort(reverse=True)
print(sub)
```

To display all the elements of the list using loop:

```
numerical techniques
DCN
DAA
TOC
data science
DAA-Lab
web technologies Lab
Python-Lab
```

The 2nd and 5th subjects are "DCN" and "data science" respectively

The first 4 subjects are

```
['numerical techniques', 'DCN', 'DAA', 'TOC']
```

The last 4 subjects are

```
['data science', 'DAA-Lab', 'web technologies Lab', 'Python-Lab']
```

True

the list after appendng COI is:

```
['numerical techniques', 'DCN', 'DAA', 'TOC', 'data science', 'DAA-Lab', 'web technologies Lab', 'Python-Lab', 'COI']
```

the list after inserting unix at pos(0) is:

```
['unix', 'numerical techniques', 'DCN', 'DAA', 'TOC', 'data science', 'DAA-Lab', 'web technologies Lab', 'Python-Lab', 'COI']
```

the list after pop() is:

```
['unix', 'numerical techniques', 'DCN', 'DAA', 'TOC', 'data science', 'DAA-Lab', 'web technologies Lab', 'Python-Lab']
```

the list after removing "unix" is:

```
['numerical techniques', 'DCN', 'DAA', 'TOC', 'data science', 'DAA-Lab', 'web technologies Lab', 'Python-Lab']
```

```
['numerical techniques', 'DCN', 'DAA', 'TOC', 'data science', 'DAA-Lab', 'web technologies Lab', 'Python-Lab', 'AD41', 'AD42', 'AD43', 'AD44', 'AD45', 'ADL46', 'ADL47', 'ADL48']
```

```
['AD41', 'AD42', 'AD43', 'AD44', 'AD45', 'ADL46', 'ADL47', 'ADL48', 'DAA', 'DAA-Lab', 'DCN', 'Python-Lab', 'TOC', 'data science', 'numerical techniques', 'web technologies Lab']
```

```
['web technologies Lab', 'numerical techniques', 'data science', 'TOC', 'Python-Lab', 'DCN', 'DAA-Lab', 'DAA', 'ADL48', 'ADL47', 'ADL46', 'AD45', 'AD44', 'AD43', 'AD42', 'AD41']
```